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**AI-BASED KPI AUDIT FRAMEWORK FOR PROMOTION
FAIRNESS IN HUMAN RESOURCES**

Management Information Systems

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Preface

This graduation project was prepared as part of the undergraduate degree requirements and focuses on the use of artificial intelligence in Human Resources decision-making processes. The study examines how AI-based models can be used not only for prediction, but also as an auditing mechanism to evaluate fairness, transparency, and consistency in employee promotion decisions.

The main purpose is to develop a KPI-based audit framework that compares objective employee performance indicators with historical promotion outcomes. The aim is to contribute to the discussion on ethical and explainable AI applications in Human Resources analytics.

I would like to express my sincere gratitude to my advisor for his guidance, feedback, and support throughout the preparation during this process.

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Abstract

The research explores the use of artificial intelligence to improve fairness and transparency in Human Resources promotion processes. The main goal of the study is to develop an AI-based auditing framework to assess the consistency of objective employee performance metrics with past promotion choices.

The study uses a large dataset of employee performance and productivity with variables such as performance score, number of projects, training hours, work hours, overtime, attendance, employee satisfaction, tenure, education level, and promotion status. These variables were grouped into four strategic Key Performance Indicator dimensions; Performance KPI, Efficiency KPI, Commitment KPI and Profile KPI. We trained a Random Forest classification model on the KPI-based fair promotion standards to predict employee readiness for promotion. Historical promotion outcomes were not used as input features in order to reduce the risk of reproducing past organizational bias.

The model outputs were compared to the actual promotion outcomes to find areas of alignment and mismatch. Employees were categorized into audit categories including correct recognition, overlooked high-potential employees, and possibly undeserved promotions. The results indicated that the model achieved better alignment on KPI-based fair datasets with accuracy around 0.77-0.80. The alignment of the model was reduced to around 0.52 when applied on real HR promotion data. This gap provides a measurable difference between being ready for promotion based on KPIs versus making promotion decisions based on history. The technical credibility and explainability of the proposed framework were further strengthened by additional validation using ROC-AUC, 5-fold cross-validation, hyperparameter tuning and SHAP-based interpretation.

The findings reveal the potential of AI-based KPI auditing systems as diagnostic mechanisms to support fairer and more transparent HR decision-making processes. Rather than substituting human judgment, the proposed framework offers a structured approach to detect potential inconsistencies in promotion decisions and to foster more evidence-based organizational practices.

Keywords: Artificial intelligence; Human Resources analytics; promotion fairness; Key Performance Indicators; Random Forest classification; explainable AI; SHAP.

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1. Introduction

In recent years, the widespread adoption of artificial intelligence and data analytics has significantly impacted Human Resources practices. Today, data-driven systems are employed by organizations to make decisions related to recruitment, performance evaluation, workforce planning and promotion processes. Such systems provide opportunities for faster, more consistent and more objective decision making. However, the use of historical organizational data is not without important concerns, especially when past decisions are embedded with subjective evaluations, hidden bias or non-meritocratic patterns.

Human Resources management has among its most important aspects the decisions of promotion, which directly influence the motivation of employees, their career development, the trust in the organization and the efficiency of the workforce. In an ideal promotion system, employees need to be assessed on quantifiable performance measures such as productivity, project completion, training attendance, attendance and commitment. However, real-life organizational promotion decisions may be influenced by other factors (not necessarily reflected in performance data) such as managerial preferences, organizational culture, interpersonal relationships or seniority considerations. This can lead to inconsistencies between the objective performance of the employee and the actual results of promotion.

In this context, AI may provide important benefits, but its design should be handled carefully. If an AI model is trained directly on historical promotion outcomes, it may reproduce existing organizational inequalities instead of correcting them. Therefore, the use of AI also should not be limited to that of a prediction tool, but it should be used as an auditing mechanism to explore whether previous decisions were consistent with objective performance data. Such an approach should allow an organisation to identify instances where high-performing employees may have been denied the opportunity to be promoted or whether promotion decisions were not substantiated by demonstrable performance indicators.

The purpose of this research is to define an AI based promotion audit framework that utilises Human Resources Key Performance Data. Utilising an expanded employee performance and productivity dataset, the research applies a Random Forest Classifier to identify employee readiness for promotion based on KPI driven data. The model is created using fair promotion standards and then assessed against actual promotion decisions. Therefore, the research will determine the disparity between KPI based promotion readiness and historical promotion decisions within an organisation's promotion process.

The primary contribution of this article is the use of artificial intelligence as a diagnostic and auditing mechanism, rather than an autonomous decision-making authority. Grouping employee-related variables into strategic key performance indicators (KPI) and performing model predictions against actual promotion outcomes creates a structured

framework through which fairness, transparency, and consistency in promotion processes can be evaluated.

The remaining content of this article is organized as follows: section 2 contains the general background literature and research on artificial intelligence, human resource analytics, performance evaluation (based on KPI), machine learning models, and fairness in decision-making algorithms; section 3 contains the material and method used by the researchers, including: the description of the data utilized, pre-processing steps, development of KPIs, model selection, training and testing procedures, evaluation metrics used, and audit framework developed for this study; section 4 contains an overview of the results of the research study, including: model performance, classification metrics (accuracy, recall/precision, etc.) interpretation of confusion matrixes, audit category results, feature importance and the impact of grouping into strategic KPIs; and section 5 includes a discussion of findings, research limitations, further research opportunities and conclusions.

This research looks at the potential for an artificial intelligence tool to identify contradictions between objective employee performance measures and prior employee advancement choices. Also, this investigation will assist to promote greater transparency, equity, and evidence-based decision-making in Human Resource management as a result of this research.

2. General Background and Literature Review

This section outlines the framework of the study from an academic and theoretical perspective. The main concepts relevant to the study include the theory of artificial intelligence (AI), machine learning, Human Resources analytics, Key Performance Indicators (KPIs), promotion fairness and evaluation of models. Also, previous studies focussing on the effectiveness of using AI in making decisions are referred to as well as having highlighted the research gap addressed in this research.

2.1 Artificial Intelligence and Machine Learning in Decision-Making

AI has become a key resource when supporting organizations' decision-making processes. Nearly every area of business uses AI to analyze vast amounts of free-form data, looking for patterns and predicting outcomes which help managers make choices. Human Resources has seen a marked increase in the use of data in the decision-making process, particularly in relation to employee evaluations and performance management; workforce forecasting and planning; and how to improve recruitment and retention rates, as well as how to consider employees for promotions and tenure.

Supervised machine learning allows models to learn from labeled data to make predictions for new, unseen data points, such as sales forecasts about new products (Kotsiantis, no date). Supervised machine learning is especially helpful for organizations to make informed decisions because of the number of variables that can affect a decision. The model is not limited to making predictions based on static rules of only variables impacting input variables; rather, it will make predictions based on the statistical relationships between multiple input and output variables.

Classification is a supervised learning task in which observations are assigned to predefined categories (Castelli, Vanneschi and Largo, 2019). In this study, promotion readiness is treated as a binary classification problem. Employee-related indicators are used as input variables, while promotion status is transformed into a binary target variable representing whether an employee is promoted or not promoted.

However, the use of machine learning in organizational decision-making also requires caution. Predictive people analytics may reproduce existing organizational patterns if models are trained directly on historical data without considering potential bias (Mezei, no date). For this reason, this study does not position AI as an automatic promotion decision-maker. Instead, AI is used as an auditing mechanism to compare KPI-based promotion readiness with historical promotion outcomes.

2.2 Human Resources Analytics

Predictive analytics in Human Resources enables organizations to uncover hidden patterns in employee data and support more data-driven workforce decisions (Alabi, 2024).

This is especially important in areas such as talent management, employee retention, performance evaluation, and workforce planning.

HR analytics can also contribute to improving employee performance and reducing turnover intention by enabling proactive and evidence-based decision-making (Bibi and Ali, 2024). In this context, employee satisfaction, training hours, productivity, work experience, and attendance can be analysed together to understand performance patterns and possible career development outcomes.

From a strategic perspective, data-driven HR systems may function as organizational resources that support long-term competitive advantage (Singh *et al.*, 2022). This perspective is relevant to the current study because promotion decisions are not only individual career decisions, but also strategic organizational decisions that affect motivation, retention, and workforce efficiency.

Promotion decisions represent a particularly important area for HR analytics because they directly influence employee motivation, organizational trust, and career development. A transparent and consistent promotion system can support employee engagement, while inconsistent promotion practices may reduce trust and create perceptions of unfairness. Therefore, analysing promotion decisions through objective indicators can help organizations evaluate whether promotion outcomes are aligned with measurable performance.

The purpose of this study is not only to predict promotion status, but also to examine whether historical promotion decisions are consistent with performance-based indicators. This audit-oriented perspective differentiates the study from traditional HR prediction studies, which usually focus mainly on improving predictive accuracy.

2.3 Key Performance Indicators in Human Resources

Key Performance Indicators can be defined as quantifiable measures used to evaluate progress toward organizational objectives (Investopedia, no date). In Human Resources, KPIs are used to assess different dimensions of employee performance, productivity, engagement, development, and retention. They help organizations transform broad performance expectations into measurable and comparable indicators.

HR KPIs play an important role in evaluating employee performance and aligning individual outcomes with organizational strategy (Gabcanova, 2012). Similarly, KPI-based performance management is not only a monitoring mechanism, but also a strategic tool that influences organizational behaviour and managerial decision-making (Setiawan and Purba, 2020).

KPI prediction can be considered both a selection and prediction problem, which highlights the importance of identifying relevant indicators and managing the model lifecycle (Thakur *et al.*, 2020). This is important for the present study because the selected KPI groups directly shape how promotion readiness is measured and interpreted.

In this study, KPIs are central to the proposed audit framework. Employee variables are grouped into strategic KPI dimensions in order to create a structured and interpretable representation of performance. The four KPI groups used in the study are Performance KPI, Efficiency KPI, Commitment KPI, and Profile KPI.

Performance KPI includes variables such as performance score, projects handled, and training hours. These indicators reflect employee output, productivity, and professional development. Efficiency KPI includes work hours, overtime hours, and attendance score, representing how employees use time and maintain work consistency. Commitment KPI includes employee satisfaction, years at company, and remote work frequency, reflecting engagement and organizational attachment. Profile KPI includes education level, age, and resignation status, representing general employee characteristics.

The use of KPI groups provides two important advantages. First, it improves interpretability by allowing the analysis to focus on broader performance dimensions instead of isolated variables. Second, it supports fairness-oriented evaluation by making the logic behind promotion readiness more transparent. Instead of producing unexplained model outputs, the framework shows which KPI dimensions contribute more strongly to promotion prediction.

2.4 AI-Driven KPI Systems and Strategic Decision Support

AI-based platforms can support decision-making by transforming structured and unstructured data into dynamic KPI insights (Khan *et al.*, 2025). This demonstrates how AI can strengthen organizational monitoring systems by making performance indicators more accessible and actionable.

AI-assisted approaches may also improve both speed and accuracy in KPI extraction compared to manual processes (Lutz *et al.*, 2023). Although this evidence comes from a maintenance context, it supports the broader argument that AI can improve KPI extraction and performance monitoring in complex organizational systems.

AI-based tools can also be used to extract Key Performance Indicators from financial reports and support benchmarking across organizations (Brito *et al.*, 2019). This finding is relevant because it shows that AI can be used not only for prediction, but also for extracting, comparing, and interpreting performance-related information.

Strategic KPI systems should also be aligned with organizational objectives. An AI-driven and collaborative framework for strategic KPI definition highlights the importance of connecting KPI systems with broader organizational goals (Corsini *et al.*, 2025). This perspective supports the structure of the current study, where employee variables are grouped into strategic KPI dimensions to evaluate promotion readiness.

Predictive analytics can support proactive performance management by shifting organizations from reactive monitoring to forward-looking decision support (Stefanovic,

2014). Similarly, AI-enabled KPI monitoring systems can be used for diagnosis and anomaly detection (Cho *et al.*, 2019). These studies support the audit-oriented perspective of the present research, where AI is used to identify inconsistencies between KPI-based expectations and historical promotion outcomes.

2.5 Promotion Fairness and Organizational Bias

Fairness in promotion decisions is an important issue in Human Resources management. Promotions are not only financial or hierarchical rewards; they also represent recognition, trust, and career progression. Therefore, promotion systems must be perceived as fair and consistent by employees.

Although AI and predictive analytics can improve strategic HR management, they also raise ethical concerns related to bias, privacy, and fairness (Al-Shammari *et al.*, 2024). This concern is especially relevant in promotion decisions because historical HR data may reflect subjective judgments or non-meritocratic patterns.

People analytics may also reproduce existing organizational patterns when historical data is used without critical evaluation (Mezei, no date). In an ideal promotion process, employees who demonstrate strong performance, high productivity, commitment, and professional development should have a higher probability of promotion. However, real organizational decisions may be influenced by factors that are not directly measurable in employee datasets. These factors may include managerial preferences, interpersonal relationships, organizational politics, seniority traditions, or informal evaluations.

Such factors do not always indicate intentional discrimination, but they may create inconsistencies between objective performance indicators and actual promotion outcomes. For example, an employee may perform strongly according to KPI-based measures but may not receive a promotion. In this study, such cases are defined as overlooked high-potential employees. Conversely, an employee may receive a promotion despite weak KPI-based readiness, which may indicate a potentially undeserved promotion from a performance-based perspective.

The aim of the proposed framework is not to prove discrimination or unfairness directly. Instead, it identifies measurable mismatches between KPI-based model predictions and actual promotion decisions. These mismatches can guide HR departments toward cases that may require further human review.

2.6 Random Forest Classification

Random Forest is an ensemble learning method that combines multiple decision trees and produces classification outputs through majority voting (Bose and Bose, 2025). This ensemble structure allows the model to reduce the limitations of individual decision trees and produce more stable predictions.

Random Forest models can improve prediction accuracy in classification and regression tasks by reducing overfitting and improving generalization (Pious *et al.*, 2024). These characteristics make Random Forest suitable for HR analytics tasks where employee performance variables may interact in non-linear ways.

Random Forest is suitable for this study for several reasons. First, it can capture non-linear relationships between employee variables and promotion outcomes. HR data often includes complex interactions between performance, efficiency, satisfaction, experience, and employee profile variables. A linear model may not fully capture these relationships, whereas Random Forest can model more flexible decision boundaries.

Second, Random Forest is relatively robust to noisy data and can perform well with tabular datasets. Since this study uses structured employee performance data, Random Forest provides an appropriate balance between predictive performance and interpretability.

Third, Random Forest provides feature importance values. This is important because the study does not only aim to predict promotion readiness, but also to understand which indicators influence model predictions. Feature importance analysis allows the study to identify whether the model relies mainly on performance-related variables or on less relevant profile-based characteristics. This supports the transparency objective of the research.

2.7 Explainability and Ethical AI in Human Resources

Explainable AI techniques such as SHAP and LIME are important for improving transparency, fairness, and accountability in predictive systems (Galanti, 2023). This is particularly relevant in HR contexts because decisions related to hiring, promotion, compensation, and termination directly affect individuals' careers.

AI-supported HR systems may improve efficiency and decision quality, but they must be carefully governed due to ethical concerns related to bias and fairness (Al-Shammari *et al.*, 2024). Therefore, explainability is not only a technical requirement, but also an ethical requirement in Human Resources analytics.

In this study, explainability is supported through strategic KPI grouping, feature importance analysis, audit categorization, and SHAP-based global interpretation. KPI grouping makes the model structure easier to interpret by organizing employee-related indicators into broader strategic dimensions. Feature importance analysis shows which variables or KPI groups influence promotion readiness, while audit categories translate model predictions into practical HR insights by identifying correct recognition, overlooked employees, and potentially undeserved promotions.

In addition, SHAP (SHapley Additive exPlanations) was implemented at the strategic KPI group level to improve model transparency. SHAP analysis provides a global explanation of how the main KPI groups contribute to the model's promotion readiness predictions. In the empirical analysis, Performance KPI had the strongest contribution, followed by Efficiency

KPI and Commitment KPI, while Profile KPI had only a limited effect. This supports the fairness-oriented logic of the study by showing that the model relies mainly on measurable work-related dimensions rather than static background characteristics.

The proposed framework therefore follows a human-in-the-loop approach. The AI model does not make final promotion decisions. Instead, it provides diagnostic evidence that HR professionals can review. This approach helps reduce the risk of blind automation and supports more responsible use of AI in organizational decision-making.

2.8 Evaluation Metrics and Audit Assessment

Classification models are evaluated by comparing predicted classes with actual class labels (Castelli, Vanneschi and Largo, 2019). In this study, standard classification metrics such as accuracy, precision, recall, and F1-score are used to assess model performance.

Accuracy measures the overall proportion of correct predictions. Precision measures how many employees predicted as promotable are actually aligned with the positive class. Recall measures how successfully the model identifies actual positive cases. The F1-score combines precision and recall into a single balanced metric. These metrics are useful because promotion prediction is formulated as a binary classification task.

Beyond these standard metrics, confusion matrix analysis is used to examine prediction errors in more detail. The confusion matrix shows true positives, true negatives, false positives, and false negatives. In the context of this audit framework, the technical error labels are interpreted through HR-specific categories. Cases where the model predicts an employee as promotion-ready but the employee was not promoted in reality are interpreted as Hidden Talent or Overlooked Employee cases. Technically, these cases correspond to false positives when historical promotion status is treated as the reference label. Conversely, cases where an employee was promoted in reality but the model does not classify the employee as promotion-ready are interpreted as Undeserved or Low Merit Promotion cases. Technically, these cases correspond to false negatives. Therefore, the audit categories provide a more meaningful HR interpretation of the confusion matrix results.

The study also uses audit category analysis to make model outputs more meaningful from an HR perspective. Employees are grouped into three categories: correct recognition, overlooked high-potential employees, and potentially undeserved promotions. This categorization transforms technical model outputs into practical organizational insights.

To strengthen model evaluation further, ROC-AUC analysis, 5-fold cross-validation, and GridSearchCV-based hyperparameter tuning were also applied. ROC-AUC assessed the model's ability to separate positive and negative classes based on predicted probabilities, while cross-validation provided a more stable estimate of model performance across different subsets of the fair training dataset. Hyperparameter tuning helped select a more suitable model configuration and reduced the risk of relying only on arbitrary model settings.

2.9 Research Gap

Existing studies on AI and HR analytics often focus on prediction accuracy, employee retention, attrition, KPI extraction, or performance forecasting. Previous research shows that predictive analytics can support HR decision-making, employee performance analysis, and strategic workforce management (Singh *et al.*, 2022; Alabi, 2024; Bibi and Ali, 2024). Other studies also demonstrate the value of AI-supported KPI extraction and performance monitoring in different organizational contexts (Brito *et al.*, 2019; Lutz *et al.*, 2023; Khan *et al.*, 2025). However, fewer studies examine the consistency between KPI-based model predictions and actual organizational promotion decisions.

This creates an important research gap. A model may achieve high accuracy in predicting historical promotions, but this does not necessarily mean that the promotion system is fair. If historical data includes biased or inconsistent decisions, a model trained directly on that data may simply reproduce existing patterns. Therefore, there is a need for audit-oriented frameworks that use AI to evaluate decision consistency rather than merely imitate past decisions.

This study addresses this gap by proposing a KPI-based AI audit framework for promotion decisions. Instead of using AI as an autonomous decision-maker, the study uses machine learning to compare objective employee performance indicators with historical promotion outcomes. By identifying overlooked employees and potentially unsupported promotions, the framework contributes to more transparent, fair, and evidence-based HR decision-making.

3. Material and Method

This portion describes the methodology utilized in the development of the study, including details surrounding the dataset, the preprocessing steps used for this study, strategic KPI construction, model selection, the process of training and testing models, metrics used for evaluation of models, and the auditor's or audit framework. The goal of this description is to detail how raw employee performance data was processed into an AI-based promotion audit system.

3.1 Data Description

As stated in this research; there was a dataset developed by Kaggle containing employee data pertaining to their behaviours while working, their productivity levels, how well they perform, and whether they were promoted or not, which will be used as the Extended Employee Performance and Productivity Data (*Employee Performance and Productivity Data*, no date).

Included within the data item, records include multiple variables such as: employee performance score, number of projects, number of training hours, average working hours per week, how many overtime hours worked, number of sick leave days, score of employee satisfaction, length of time the employee has been with the company, number of times working from home, employee education level, employee age, whether the employee resigned, and whether the employee received a promotion.

These items were chosen because they include many different types of information about employee performance and behaviours, along with including representative data from multiple areas of the organization's operational performance management framework.

The types of variables related to performance would include: Performance score (i.e., the employee's current performance level), number of completed projects, number of completed hours of training, etc., which measure output and employee development. The types of variables measuring how efficiently employees are working are: Average work hours, overtime hours, attendance, etc. They capture how employees are managing their time and performing consistently from period to period.

The types of variables related to employees' commitment to the organization are: employee satisfaction score and time at the company, which measure engagement and are assessed over time for individual employees in relation to the organization.

The types of profile-based variables selected are: age, education level, and resignation status, which provide an additional dimension of employee background that could assist with understanding the relationship between variables and the outcome variable.

The outcome of interest for this analysis is the Promotion variable. The Promotion variable has been transformed to two classes (Promotion and No Promotion), as the

promotion variable is a binary variable. Any employees that received a promotion have been classified as a Promotion, and any employee that has not received a promotion has been classified as No Promotion; thus, the problem can be formulated as a binary classification problem.

3.2 Data Preprocessing

To ensure the data set is ready for machine-learning analysis, multiple pre-processing steps were carried out before training the model. The first task was to convert all categorical variables into numeric representations. An example of a categorical variable that was assigned numeric coding is education level. Education levels were ordered from High School to Bachelor to Master to PhD and assigned increasing numeric values for each category. As a result, the algorithm will be able to recognize the level of education of an employee, and will treat them as comparable with each other based on their educational background.

Another categorical variable that was assigned numeric coding is resignation status (yes or no). This allows the use of resignation information to compare to other numeric employee attributes.

A third variable that was manipulated for additional analysis is sick days. A new variable was created from sick days, called attendance score. Since a higher number of sick days indicates lower attendance consistency, the sick days variable was reversed to create an attendance score. In this way, employees with fewer sick days received higher attendance scores.

The last part of the pre-processing steps were to handle any missing values. The method used to fill any missing values was the median of the column's values. This method is preferred over the mean because the median is less impacted by extreme data points compared to the mean. All of these pre-processing steps were performed to get the data into a clean format for building KPIs and training.

3.3 Strategic KPI Construction

The study made an important methodological advancement by converting raw employee data into groups of Strategic Key performance indicators (KPIs). Rather than using individual employee data (or variables) for analysis purposes, we grouped related variables into one of four Strategic KPI dimensions - thereby providing an improved way to interpret model results through a structured human resource management (HRM) viewpoint.

The Strategic KPI groups established in the study are connected to employee performance (Performance KPI), efficiency of employees (Efficiency KPI), commitment to employees/the organization (Commitment KPI), and profile of employees (Profile KPI).

The Performance KPI contains the following metrics: employee performance score (i.e., performance appraisal), number of projects completed, and number of hours in training.

Overall, this performance KPI category is an indicator of employee output, productivity and professional development.

The Efficiency KPI contains the following metrics: average number of weekly work hours (standard and overtime), and employee attendance score. The efficiency KPI is a measure of time management, work intensity and consistency in attendance, respectively, of employees.

The Commitment KPI contains the following metrics; employee level of satisfaction with employment, number of years with employer, and number of days working from home. Overall, this commitment KPI category is a measure of employee commitment to their job or organization, whether they have a connection with their organization, and patterns in where they work.

The Profile KPI contains the following metrics; education level (high school, college, etc.), age, and employee resignation status. The Profile KPI category represents the background information about the employees.

Variables measured within the performance KPI group had been measured on different scales, therefore in order to develop KPI group scores each variable was scaled using a min-max (maximum and minimum) normalization system. By utilizing the min/ max normalization method, all data was converted into comparable ranges. Subsequently, all normalized employee variables measured within each KPI grouping were averaged together measuring employee attributes defined by the KPI category. Therefore, each employee was assigned to 4 strategic KPI groups (Performance KPI; Efficiency KPI; Commitment KPI; and Profile).

This structure made the model easier to interpret because promotion readiness could be analysed through broader performance dimensions rather than isolated raw variables.

3.4 Model Selection

The study chose the Random Forest classification algorithm to be the primary machine learning method. Random Forest is an ensemble learning technique that produces a more stable and accurate prediction by combining several decision trees. This method is helpful for this study because HR data can contain non-linear relationships between indicators of an employee's performance and their likelihood of being promoted.

The Random Forest method was chosen for three primary reasons; it can model different interactions among KPI groups regardless of whether there is a linear relationship or not, it is less sensitive to noisy data than a single decision tree and therefore avoids over-fitting quite as much, and finally, the feature importance output from the model allows us to make sense of the model and provides insight into which KPI groups are most strongly associated with readiness to be promoted. As such; the study's goal is not only to predict who will be promoted but also evaluate whether or not promotions have been applied consistently;

thus the model needs to be able to support auditing processes, and the feature importance output from the Random Forest model allows us to ascertain which KPI groups are strongly contributing to the prediction of readiness for promotion.

3.5 Model Training and Testing

The model training process was developed to determine whether employees could receive a promotion based on a fair standard and a KPI-driven logic. Rather than directly training the model using all historical results of promotions, a fair promotion dataset was created that included only those promoted based on KPI-based measures and constraining to historical objective promotion outcomes. Creating the fair promotion dataset to evaluate the model's performance was preferred because the historical HR promotion data are subjectively determined and have both organizational and subjective factors that are therefore not measurable by comparison to KPI-based actions. If training were conducted against all historical HR promotion outcomes, the conclusion would be the reproduction of many of the HR inconsistencies rather than offering an independent audit-like evaluation.

The input for the model was based on the four (4) strategic KPI dimensions (1) Performance KPI; (2) Efficiency KPI; (3) Commitment KPI; (4) Profile KPI. Each of the KPI dimensions was computed based on the average normalized values for variables of strategic importance included within the KPI dimension, and no additional weighting factor was introduced within KPI dimensions or across KPI dimensions. Thus, for the purpose of constructing logic for promotion readiness, all variables within each KPI dimension were treated the same and all KPI dimensions were treated equally in relation to the logic structure.

The calculation of composite scores for each KPI dimension has been provided as follows:

$$\text{KPI Group Score} = \text{Sum of normalized variables within the KPI group} / \text{Number of variables within the KPI group}$$

After the four KPI group scores were calculated, an overall KPI readiness score was created for each employee:

$$\text{Overall KPI Readiness} = (\text{Performance KPI} + \text{Efficiency KPI} + \text{Commitment KPI} + \text{Profile KPI}) / 4$$

The overall readiness score was established as a formal indicator of measurable promotion readiness. No predefined absolute threshold such as 0.70 or 0.80 was levied on the research study. The fair promotion dataset was then made up of historical promotion results that were generally aligned with an employee's overall KPI-based readiness profile. Specifically, KPI-based promoted and non-promoted employees were used as KPI-consistent cases.

To identify KPI-consistent cases operationally, employees were evaluated using overall KPI readiness scores and the promotion outcome distribution to identify KPI-consistent promotion. The KPI-consistent promotion criteria were to identify employees with both high overall KPI readiness and the promotion outcome and to include employees with both low overall KPI readiness and no promotion outcome.

The objective of this filtering process was not to label the excluded observations as unfair. Rather, it was to build a merit-consistent training subset in which promotion outcomes were more clearly supported by measurable KPI indicators. This allowed the Random Forest model to learn a performance-oriented promotion standard before being applied to the full real HR dataset as an audit tool.

The Random Forest model was trained on this fair promotion dataset. The binary outcome for the target variable (the promotion status) had a binary coding scheme where an employee with an open promotion would be given a value of 1, while an employee who has never been promoted would be coded as a 0. The initial Random Forest model was trained with 100 estimators and a random seed value, in order to ensure replicability. Upon completion of training the Random Forest model with 100 estimators, hyperparameter tuning was completed using grid search (GridSearchCV) to identify hyperparameters that resulted in the highest cross-validated weighted F1 score.

The model was evaluated using two separate testing stages after its completion, with the first stage quantifying the model against KPI-aligned synthetic datasets, which measured the promotion logic of the model in a controlled and measurable way. The second test involved applying the final version of the Random Forest model to the entire dataset of HR records from each employee in the population. This second stage was critical to the audit purpose of the study because it allowed KPI-based model predictions to be compared with historical promotion outcomes.

The approach used is explained with this flowchart:

Raw employee dataset → KPI construction → KPI-consistency filtering → fair promotion training dataset → Random Forest training → synthetic dataset validation → real HR audit comparison

This design provides a way to tell apart two research questions: the ability of this model to successfully learn and apply a quantifiable KPI-based standard for determining promotions and the degree to which real-world promotion decisions (made on an earlier dataset) matched up with the one developed from the model. As a result, the model was intended not to influence promotion decisions but to serve as a means of reviewing the degree of alignment (or lack thereof) between the two.

3.6 Evaluation Metrics

To determine the model's predictive performance and relevance to auditing, various classification metrics were utilized. The most significant metrics included accuracy, weighted precision, weighted recall, weighted F1-score, ROC-AUC, cross-validation results, hyperparameter tuning results, and analysis of the confusion matrix.

Accuracy is defined as the overall proportion of observations that were classified correctly. Weighted precision is the extent to which a model can be trusted to assign employees to either promotion or non-promotion classes. Weighted recall is the measure of success for a model in identifying true class outcomes for both types of employees. The weighted F1-score combines precision and recall into one value and is particularly useful in evaluating promotion audit results because both false positives and false negatives must be considered.

Additionally, the ROC-AUC was included to evaluate how well the model could separate promotion from non-promotion classes using predicted probabilities. In contrast to accuracy, the ROC-AUC calculates a score with less sensitivity to a single classification threshold and gives additional insight into the model's usefulness as a classification tool.

5-fold stratified cross-validation using an appropriate KPI-consistent fair training dataset was applied to check for model stability. Each fold retains the same distribution of each class. These allow the model to be trained and evaluated on independent subsets of data.

GridSearchCV was used to tune the hyperparameters for the model, allowing for the evaluation of multiple combinations of Random Forest hyperparameters (number of estimators, maximum tree depth, minimum samples needed to split, and minimum samples per leaf) to find the best performing model. The best performing model will be the one producing the highest cross-validated weighted F1-score.

A confusion matrix was used to determine the type of errors produced by the model. This will help with the evaluation of the errors produced for the audit framework because different types of errors relate to different HR meanings. For example, if the model produces an incorrect prediction where an employee is predicted to be promotion-ready and subsequently the employee was not promoted, this would indicate there may be a hidden talent/employee that was overlooked. On the other hand, if an employee was promoted and the model predicted that employee was not promotion-ready, this would indicate there may be a potentially unsupported promotion (with respect to the original KPI indicators).

To enhance the reliability of the assessment, the model was compared on various datasets. It was first validated through cross-validation against the KPI-aligned fair training dataset. Then, the final tuned model was applied to the complete real HR data set. This comparison provided a way to determine if the model could accurately predict promotions using KPI-based logic under fair conditions and to assess whether the actual promotion history aligned with this logic.

Thus, the evaluation not only attempted to determine how well the model predicted promotions, but it also established that any model errors produced by this validation process could be considered to have significance as a result of their use as audit signals in evaluating fairness of HR promotion practices.

3.7 Audit Framework

In addition to providing insights into trending predictive analytics and helping to evaluate the closeness between KPI Based Promotion Readiness and Actual Promotion Outcome, the AI Based Audit Framework introduced by this Data Science project has an additional purpose of identifying any employees whose promotions should be flagged for further consideration by Human Resources but does not make a final determination regarding the promotion of a given employee. Employees are divided into three distinct categories based on the audit framework:

Correct Recognition: The model prediction and actual promotion outcome are aligned. This means that the employee's KPI-based readiness is consistent with the historical promotion decision.

Hidden Talent / Overlooked Employee: The model predicts the employee as promotion-ready, but the employee was not promoted in reality. These cases may represent high-potential employees who were not recognized by the historical promotion process.

Undeserved / Low Merit Promotion: The employee was promoted in reality, but the model does not predict promotion readiness. These cases may represent promotion decisions that are not strongly supported by the selected KPI indicators.

This categorization transforms technical model outputs into practical HR audit insights. By identifying alignment and mismatch cases, the proposed framework supports a more transparent and evidence-based review of promotion decisions.

3.8 Dashboard Prototype

A prototype of a Streamlit dashboard was developed alongside a machine learning model to exemplify how the audit framework can be operationalized. The dashboard is an interactive interface that allows users to upload an employee dataset, search for an employee by Employee ID and review the promotion recommendation given by the model and the strategic KPI scores for that employee.

The screenshot of the dashboard included below (Figure 1) shows the employee recommended for promotion based on the predictions of the Random Forest model; provides a confidence score for the suitability for promotion; and provides a graphical representation of the selected employee's KPI profile in a bar graph. In addition, the dashboard provides an alternative pie chart display of the relative distribution of KPI scores. Lastly, the raw KPI

values are available in a table format to facilitate the user noticing the employee's Performance KPI, Efficiency KPI, Commitment KPI and Profile KPI scores easily.

It should also be noted that the datasets provided for this study don't contain a field for the name of the employee; therefore the dashboard displayed in the screenshot above indicates "Unknown Employee". If a user's uploaded dataset contains the employee name field, the dashboard will display the employee's name along with the Employee ID.

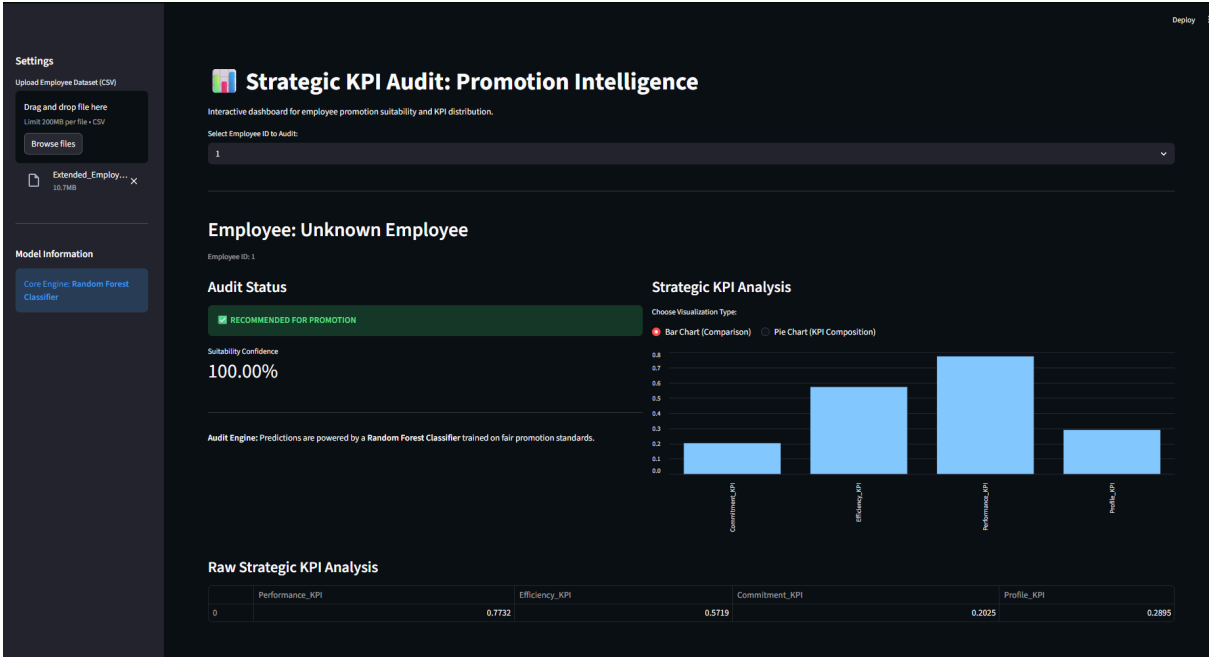


Figure 1. Streamlit Dashboard Prototype for Employee-Level Promotion Audit

HR decision-makers will not be replaced by this prototype. This prototype demonstrates the possible application of the AI-based audit framework as an interactive and usable decision-support system. This will give HR professionals the ability to assess their employee promotion readiness on a per-employee basis, as well as allow them to evaluate the strategic KPI dimensions used in evaluating employees and identify employees needing additional evaluation before promotion.

4. Findings

The results/findings from the AI-based Promotion Audit Framework occur in this section. Each of the results is organized into model performance, classification metrics, confusion matrix analyses, audit results by category/feature, feature importance, KPI group strategic effects, and comparisons of KPI Fair Data to HR Real Data. The intent of this section is to illustrate what the model found and how the results of the audit support the objectives of the study.

4.1 Model Performance Evaluation

Stage 1 of the analysis examines the total performance of the Random Forest Classification Model. The Random Forest Classification Model has been trained using KPI Fair Promotion Standards and subsequently assessed using diverse datasets to determine whether or not there is a strategic KPI that represents the candidates' readiness for promotion.

Table 1. Cross-Validation and Hyperparameter Tuning Results on the Fair Training Dataset

Evaluation Component	Result
5-Fold CV Accuracy	0.9651 ± 0.0022
5-Fold CV Weighted F1-Score	0.9651 ± 0.0022
5-Fold CV ROC-AUC	0.9935 ± 0.0009
Best Cross-Validated Weighted F1-Score	0.9674
Best Parameters	max_depth = 10, min_samples_leaf = 2, min_samples_split = 2, n_estimators = 100

The Random Forest model demonstrated excellent cross-validation results on the KPI-consistent fair training dataset, as reported in Table 1. The model achieved a 5-fold cross-validation accuracy of 0.9651 and a weighted F1 score of 0.9651; the ROC-AUC score was also high at 0.9935. These high accuracy metrics indicate that the Random Forest model has a high level of consistency in learning the KPI-consistent promotion hierarchy across multiple data folds. During hyperparameter tuning of the model, the following values were chosen for the Random Forest settings: Maximum depth of 10, Minimum samples per leaf of 2, Minimum samples split of 2, Number of estimations (forests) of 100. After tuning the model, the hyperparameter tuning yielded a 5-fold cross-validated weighted F1 score of 0.9674, indicating that the tuned Random Forest model provided a slightly improved and more stabilised fit to the data than did the baseline model. Overall, the high levels of accuracy achieved through these data processes should only be viewed as potentially indicative of how well the model(s) will perform when using historical HR decisions as the basis for comparison to recent HR decisions. While the results reported suggest that the Random Forest

model is capable of learning a stable KPI-defined promotion hierarchy before being used for audit comparisons, caution must be exercised when interpreting any of these two models' high levels of cross-validated accuracy on the KPI-consistent fair training dataset.

The model performance was evaluated using accuracy, weighted precision, weighted recall, and weighted F1-score. These metrics were selected because promotion prediction was formulated as a binary classification task, where employees were classified as either “Promotion” or “No Promotion.” For clarity, the main performance metrics were summarized in a single comparative table.

Table 2. Model Performance Metrics Across Datasets

Dataset Type	Overall Accuracy	ROC-AUC	Weighted Precision	Weighted Recall	Weighted F1-Score
Synthetic KPI Dataset 1	0.7682	0.9255	0.83	0.77	0.77
Synthetic KPI Dataset 2	0.8013	0.9299	0.84	0.80	0.80
Real HR Dataset	0.5159	0.5039	0.56	0.52	0.53

The results in Table 2 reveal that the tuned model performed significantly better on the two synthetic KPI-aligned datasets than it did on the full real HR dataset. On Synthetic KPI Dataset 1, the tuned model achieved an accuracy of 0.7682 and a ROC-AUC score of 0.9255. On Synthetic KPI Dataset 2, it achieved an accuracy of 0.8013 and a ROC-AUC score of 0.9299. The ROC-AUC values indicate that the model was able to distinguish between promotion and non-promotion classes effectively when promotion outcomes followed measurable KPI-based patterns.

In contrast, the accuracy on the real HR dataset was only 0.5159 and the AUC was only 0.5039 for the tuned model. Since an AUC of near 0.50 indicates that the promotion outcomes are not highly separable, this suggests that the historical promotion outcomes in the real dataset were not highly separable according to the KPI-based promotion readiness standard learned from the KPI-consistent fair training dataset.

Therefore, the difference between the performance of the tuned model on the synthetic datasets and the real dataset is not strictly a model performance issue, but it does indicate a measurable deficiency between the KPI-consistent logic of promotion decisions and the historical HR promotion outcomes. This deficiency in the alignment of the KPI-consistent baseline for promotion decisions and the HR historical promotion outcomes is the primary

rationale for the introduction of the audit framework, as it identifies where a further human review may be warranted.

4.2 Classification Metrics Analysis

The classification metrics provide a more detailed view of how the Random Forest model performed across different datasets. While accuracy provides an overall measure of overall agreement/ alignment, weighted precision, weighted recall, and weighted F1-score provide a more balanced approach to evaluating the model's classification performance across both classes (promotion vs. non-promotion). Since these weighted metrics aggregate class performance while taking the distribution of instances across classes into account; therefore, they may provide additional decision support in weighing overall model results.

Based on presentable information from Table 2, the Random Forest was much more stable when modelling KPI-aligned synthetic datasets versus the actual HR data. The weighted precision values were 0.83 and 0.84 for the two synthetic datasets, while the weighted F1-score values were 0.77 and 0.80. These results indicate that more consistent classification results were achieved with the Random Forest model when the promotion labels were in alignment with KPIs.

The ROC-AUC results back up this interpretation. For KPI-aligned synthetic datasets, ROC-AUC scores of 0.9255 and 0.9299 suggest much higher separability between promotion and non-promotion classes than found with true historical promotion outcomes in the HR dataset, where the ROC-AUC score was 0.5039 (i.e., random).

In the real HR dataset, weighted precision decreased to 0.56 and weighted F1-score decreased to 0.53. This weaker performance suggests that real historical promotion outcomes were not strongly aligned with the KPI-based promotion logic learned by the model. This suggests that the real HR dataset includes more mismatched cases where historical promotion outcomes do not fully align with the KPI-based model predictions. Overall, audit-type metrics indicate that the promotion logic developed from KPI definition can be learned consistently in an unbiased setting while the historical (real) promotion data is not consistent with KPI measures.

4.3 Confusion Matrix Analysis

Using the confusion matrix (see Figure 2), we were able to analyze the types of correct and incorrect model predictions. The confusion matrix gives a more detailed view than the overall accuracy of how, in terms of their actual outcome, the actual promotion decisions were made based on KPI-based predictions using the Random Forest model.

The confusion matrix used in this study contains actual results for promotion decisions and the promotion outcomes predicted by the Random Forest model. The diagonal cells of the confusion matrix represent those instances when the predicted promotion from the Random Forest model were consistent with the historical promotion decision made for each employee.

The off-diagonal cells of the confusion matrix represent instances when there was a mismatch between the predicted promotion and the historical promotion decision made for each employee.

As clearly illustrated in Figure 2, the real HR dataset consists of records of both aligned and mismatched cases. Identifying all of the employees with mismatched promotion predictions is crucial for this audit framework because these records identify employees whose historical promotion decisions do not fully align with the predicted results based on KPI-based readiness.

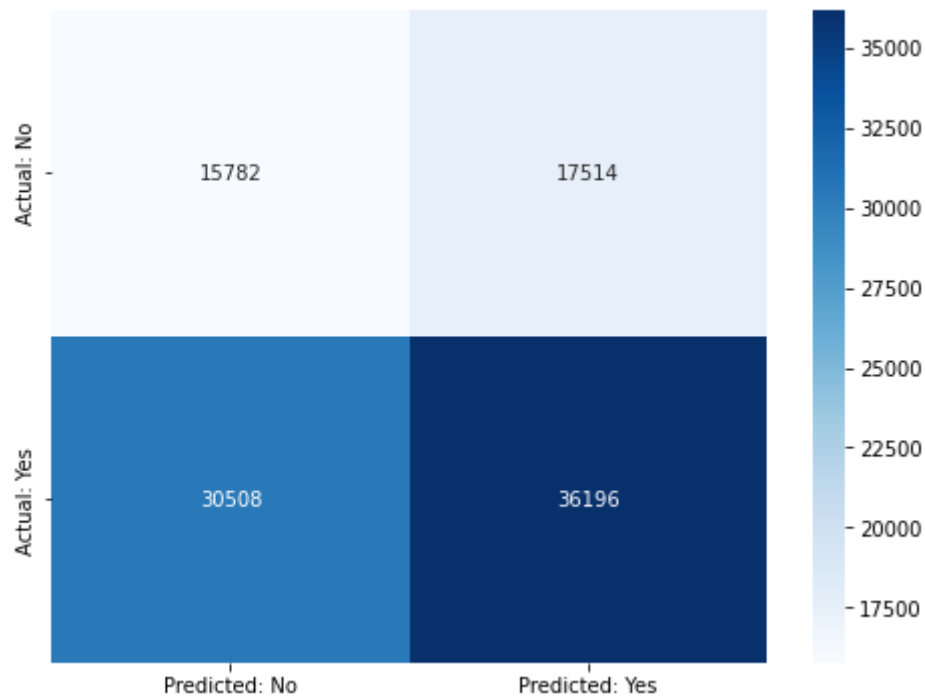


Figure 2. Confusion Matrix: KPI Logic versus Actual Promotion Decisions

In terms of an audit, the cases where the model predicts that an employee is promotion-ready, but the employee was not promoted is considered an Hidden Talent or Overlooked Employee case. This means that the employee may have measurable KPI performance which indicates that they are promotion-eligible based on measured performance, however, the employee's historical promotion outcome does not support this potential. Technically, these types of cases are false positives based on using actual historical promotion as the reference label.

In the reverse case, the employee was promoted in real life but the model does not classify him/her as promotion-ready, this is an Undeserved or Low Merit Promotion case. This means that the promotion may not be supported by the selected KPI indicators. Technically speaking, these cases would be classified as false negatives based on actual historical promotions as the reference label.

Thus, the confusion matrix supports the primary audit objective of the study. It does more than indicate whether the model is correct or not; it is useful for identifying where the alignment between objective performance indicators and actual promotion decisions is breaking down.

4.4 Audit Category Results

To improve the ability for HR auditors to interpret the model results, employees were assigned to audit categories based on how well their actual promotions matched the KPIs from the model. This categorization allows for the conversion of technical prediction results to applicable HR insights.

The audit framework includes three primary categories. Correct Recognition refers to employees whose actual promotion outcome matches the model-based prediction. Hidden Talent, or Overlooked Employees, refers to employees who were predicted as promotion-ready by the model but were not promoted. Undeserved or Low Merit Promotion refers to employees who were promoted in reality but were not classified as promotion-ready by the model.

The pattern of the Synthetic KPI datasets shows a much higher concentration of Correct Recognition (see Figures 3 and 4). The results of which align with our expectations, in that the construction of these datasets was generated based on the KPI-based fair promotion rationale. The predictive model and the actual promotions would therefore be more consistent with each other under a synthetic fair circumstance.

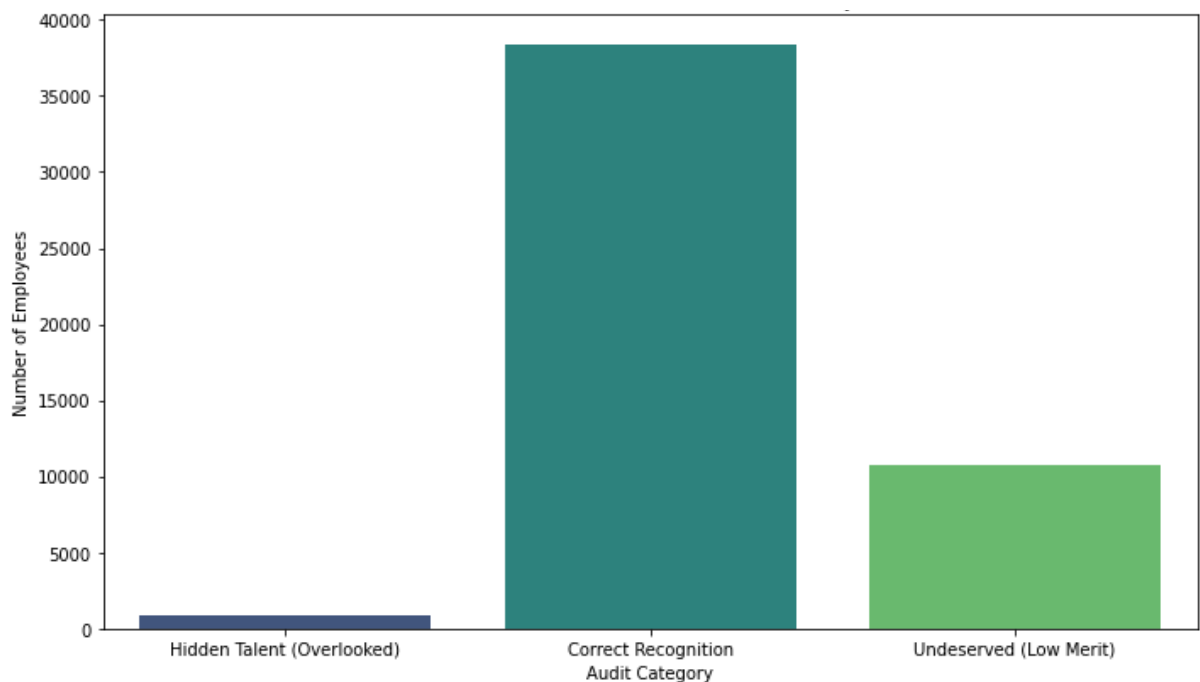


Figure 3. Audit Category Distribution for Synthetic KPI Dataset 1

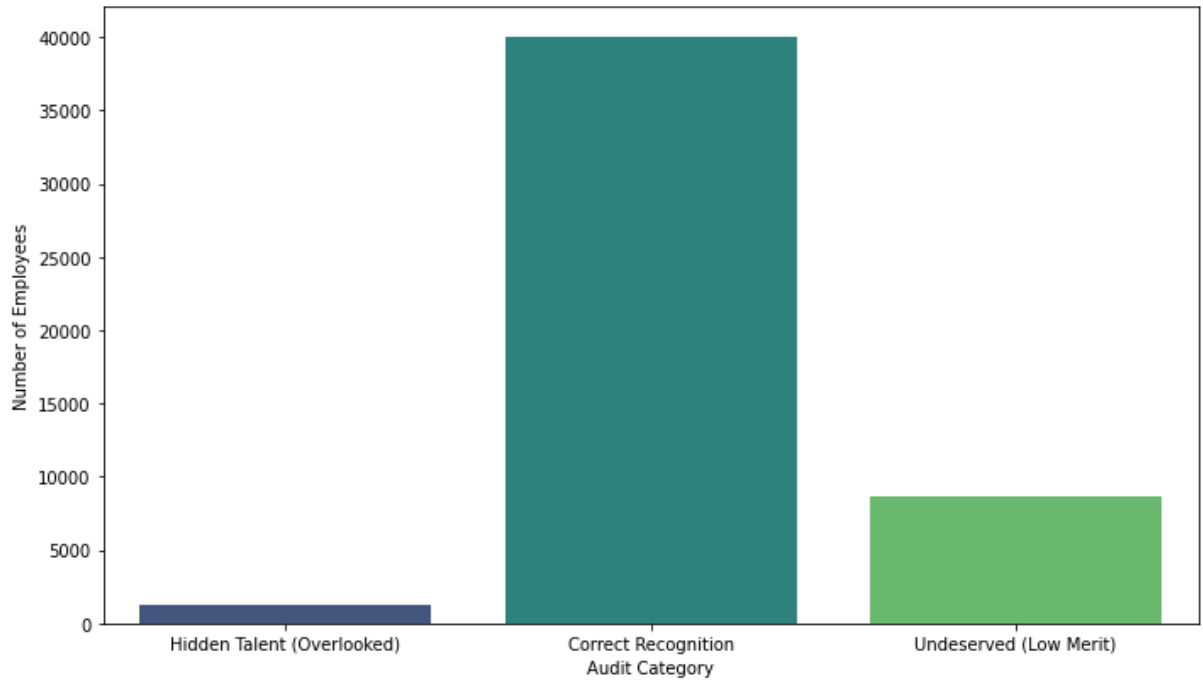


Figure 4. Audit Category Distribution for Synthetic KPI Dataset 2

In contrast, Figure 5 shows the audit category distribution for the real HR dataset. Compared with the synthetic datasets, the real dataset includes a more visible number of Hidden Talent and Undeserved Promotion cases. This indicates that historical promotion decisions do not fully follow the KPI-based promotion readiness logic learned by the model.

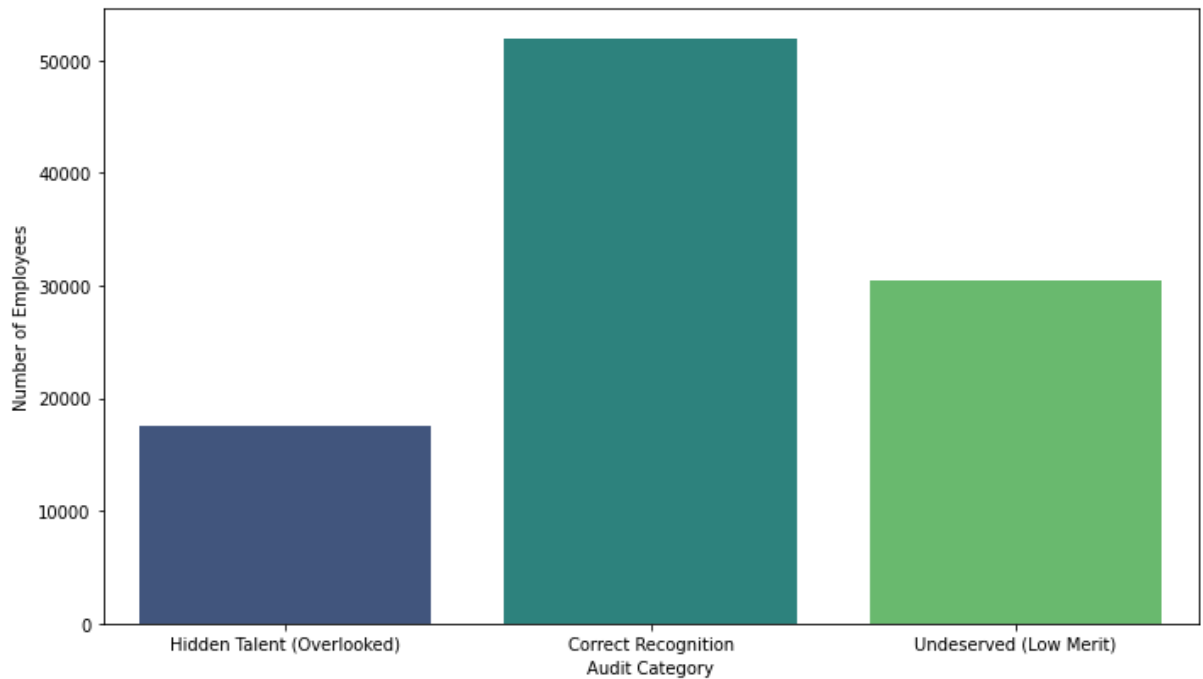


Figure 5. Audit Category Distribution for Real HR Dataset

The Hidden Talent Classification is an important part of the HR review process that involves identifying employees who have been classified as promotion ready according to KPI-based performance indicators but may not have been promoted in previous historical data sets. Employees in the Hidden Talent Classification can represent previously overlooked high potential individuals who have not had their measurable performance fully recognized.

The Undeserved Promotion Classification includes employees who have been promoted in the real world but do not meet the model's definition of Promotion Ready. These employees do not necessarily represent actual cases of unfairness, but they do illustrate that some promotion decisions may have been made using information outside of the selected KPI indicator.

The audit category analysis overall provides evidence that supports the main purpose of this study by identifying specific groups of employees whose promotion outcomes should be reviewed in further detail. The comparison between the synthetic and real data sets demonstrates that this recommended AI-based framework provides a diagnostic tool for evaluating promotion consistency; it does not function as an automated decision-making tool.

4.5 Supplementary Feature Importance Analysis

Along with the strategic KPI group-level interpretation, a supplementary raw-feature importance analysis was conducted to examine how each original employee indicator related to promotion readiness. This analysis should be considered as supporting descriptive evidence rather than the primary model explanation, because the main model was structured around four strategic KPI groups: Performance KPI, Efficiency KPI, Commitment KPI, and Profile KPI.

The results of this analysis, which are illustrated in Figure 6, indicate that the most important single indicators are performance score, attendance score, number of projects handled, training hours, and employee satisfaction score. These indicators represent employee output, attendance, development, and commitment. In comparison to these variables, static/background type variables show relatively less influence.

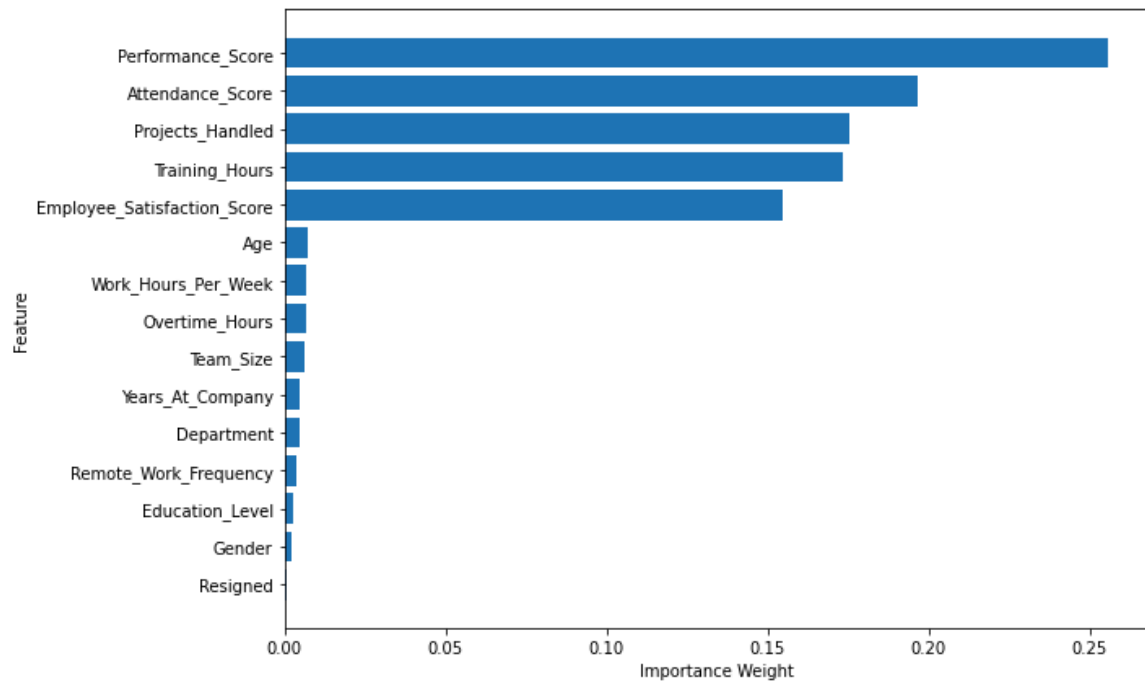


Figure 6. Supplementary Raw-Feature Importance Results

These patterns are consistent with the guidelines of the KPI-based auditing framework. Promotion readiness is more closely linked to objective, quantifiable performance indicators than to arbitrary or fixed profile traits. Since the main model interpretation is conducted at the strategic KPI group level, the raw-feature importance results should be interpreted only as supplementary evidence supporting the primary KPI-based model explanation.

4.6 Strategic KPI Group Effects and SHAP-Based Interpretation

The study analyzed not only the raw feature importance but also how the larger strategic KPI groups related to the model's prediction of promotion readiness. This gives the data a more organized interpretation as the original employee variables were categorized into four primary dimensions: Performance KPI, Efficiency KPI, Commitment KPI, and Profile KPI.

From the figure below (Figure 7), it can be concluded that the group of Performance KPIs has the greatest relative impact on the model's predictions. Performance KPIs include performance scores, number of projects completed, and training. Its leading position suggests that employee output, productivity, and professional development are the most influential dimensions in the model's assessment.

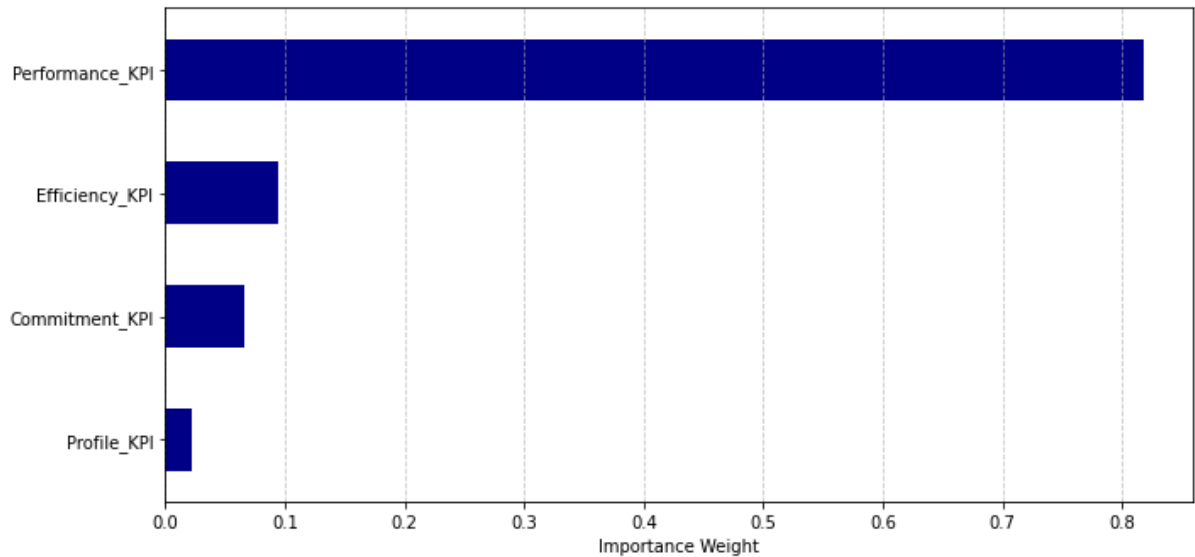


Figure 7. Impact of Strategic KPI Groups on Promotion Decisions

The Efficiency KPI group also contributes meaningfully to the prediction process. Since this group includes work hours per week, overtime hours, and attendance score, its influence shows that time management, work intensity, and attendance consistency are relevant for evaluating promotion readiness.

The Commitment KPI group has a moderate influence on the prediction process. This group includes employee satisfaction score, years at company, and remote work frequency. These variables reflect engagement and organizational attachment, which may support promotion readiness but appear to have a lower influence than direct performance and efficiency indicators.

The Profile KPI group has the lowest measure of influence from the four areas being evaluated. Since the constituents of this KPI group are education, age, and whether or not an employee has ever resigned from the company, their lower contribution indicates that this project's model used very little of the static descriptive characteristics of employees to determine promotion readiness. This reflects the study's fairness-based objective to assess promotion readiness based on measurable, work-related criteria rather than based on the applicant's personal attributes.

The strategic KPI group analysis provides the main model-level interpretation of the proposed audit framework. While Figure 6 offers supplementary insight into the relevance of original employee indicators, Figure 7 shows how these indicators are represented within broader HR performance dimensions. Together, these analyses improve interpretability, but the primary explanation of the model is based on the strategic KPI group structure.

To improve model interpretability further, SHAP analysis was also conducted at the strategic KPI group level. As shown in Figure 8, Performance had the largest mean absolute SHAP value of (0.3763) followed by Efficiency (0.0785) and Commitment (0.0561) KPI

groups with Profile KPI having a very minor contribution (0.0029). These findings confirm the strategic importance of each KPI group and add to the evidence provided by SHAP by determining how significant the contribution of each KPI group to the model is regarding the prediction of promotion readiness.

The SHAP findings are very much reflective of the fairness-focused structure of the proposed Audit Framework model. This model predominantly uses metrics related to measurable contributions of employees and work-related factors, specifically performance-based factors, compared with static characteristics of the employee profile; therefore, the SHAP findings substantiate the rationale that this model's decision logic aligns with principles of transparent and performance based HR evaluations.

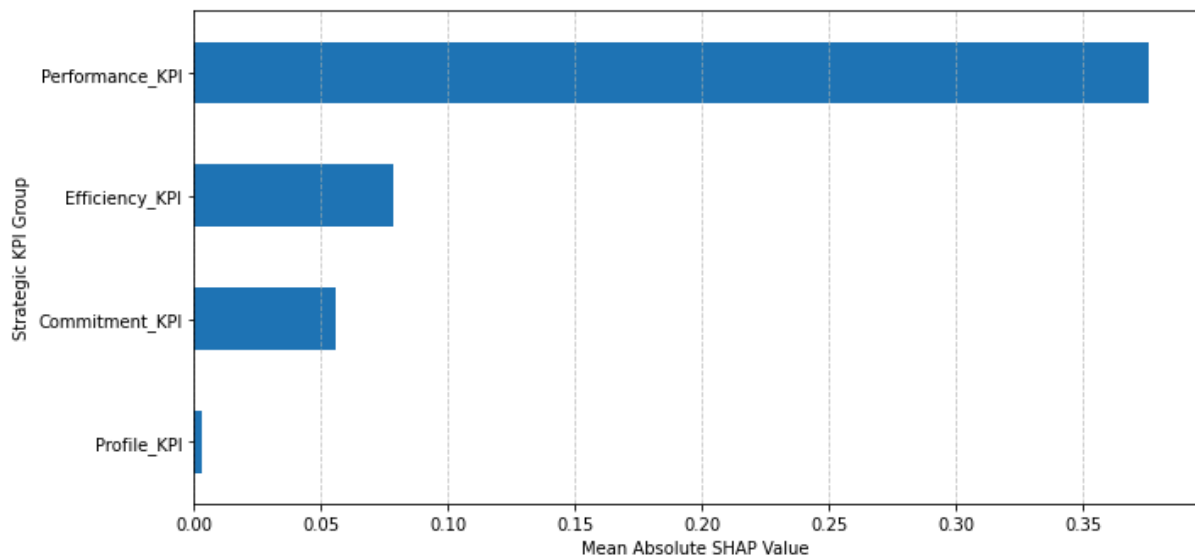


Figure 8. SHAP Global Importance of Strategic KPI Groups

4.7 Comparison Between KPI-Based Data and Real HR Data

A key finding of this research has been the contrast between the KPI-oriented datasets and the real HR dataset. The KPI-defined synthetic datasets appear to represent promotion conditions characterised by fairness, with promotion outcomes determined based on performance metrics. However, the real HR dataset encompasses an organisation's historical promotion decisions which could incorporate both KPIs as well as other considerations related to promotion.

Looking at Table 3, we see the model was able to achieve closer alignment to the KPI-oriented datasets with the accuracy values being 0.77 and 0.80, respectively, while the ROC-AUC values were above 0.92. However, when the same model was run against the real HR dataset, the model demonstrated an alignment accuracy of 0.52. The implication of this is that real-world promotion decisions cannot be fully represented by the KPI-based promotion logic learned by the model.

Table 3. Comparison Between KPI-Based Data and Real HR Data

Comparison Area	KPI-Based Synthetic Datasets	Real HR Dataset
Promotion logic	Based on measurable KPI indicators	Based on historical promotion outcomes
Model alignment	Higher, approximately 0.77-0.80	Lower, approximately 0.52
Classification performance	More balanced and consistent	Weaker and less aligned
Audit category distribution	More cases fall under correct recognition	More visible mismatch cases
Interpretation	Model successfully learns KPI-based fair logic	Historical decisions do not fully match KPI-based logic
Audit meaning	Confirms model validity under fair conditions	Reveals potential cases for HR review

According to the analysis performed by the AI model trained in accordance with the KPI (Key Performance Indicator) standard for promotion, it performs consistently when using KPI-based standardised fair promotions than when using non-KPI standardised historical promotion data. The reason for the difference has to do with the similarity of the training dataset to KPI-based promotions as compared to non-KPI promotions found in the real HR Dataset. The implication is that with proper documentation, the KPI structure represents promotion eligibility from a strategic perspective, under controlled and measurable conditions.

The poor alignment of KPI-based promotions with the real HR Dataset does not represent a limitation of the AI model, but rather is significant for the purpose of auditing this research. Because the AI model was trained to learn logic for merit-based promotions through the use of KPIs, its poor alignment to real promotions suggests that in many cases, historical promotion outcomes may be based on factors that are not easily measured through performance indicators.

Therefore, the comparison of KPI-based promotions as compared to Real HR Data supports the research's primary findings. AI-based audits can be of considerable use in identifying the areas where past promotion decisions were in compliance with objective performance criteria as well as in identifying where decisions may need additional evaluation.

4.8 Summary of Findings

According to what this study found, Random Forest is better at predicting outcomes related to merit-based promotions when the metrics being used to evaluate merit-based solutions are measurable based on KPIs. The Random Forest model received accuracy ratings of ~ 0.77 and 0.80 for 2 synthetic datasets analyzed by the study. These results suggest that the Random Forest model was able to learn the promotion pattern when decisions were structured around measurable employee performance indicators.

However, upon testing the same model with the entire real HR database (not only the filtered database used to create the synthetic datasets), accuracy significantly decreased (to ~ 0.52). This result suggests a measurable difference between merit-aligned promotion standards and broader historical promotion outcomes. Moreover, the decrease in accuracy indicates that the original database still contains examples of employees who were not promoted for reasons that cannot be explained through common KPI metrics.

Upon evaluation of the confusion matrix and audit categorization, it is evident that not only were there examples of firmer-supported promotion decisions, there were instances of employee promotion-related eligibility being misaligned with promotion decision-making criteria. Hidden Talent cases (examples of employees eligible for promotion based on performance indicators but not actually promoted) and Undeserved or Low Merit Promotion cases (examples of employees being promoted but not deemed eligible by the model) fall within the respective confusion matrix and audit categorization. While neither of these measures definitively demonstrate an unfairness in the employee promotion process, these categories identify cases that may benefit from further HR review.

Results of the raw feature analysis of promotional potential support the relationship between work-related indicators (such as performance score, projects managed, hours of training, attendance score, and employee satisfaction) with promotion potential. Performance and efficiency had the greatest influence on promotional predictions while profile had relatively little influence based upon KPI group based analysis of the relevance assignment. This is consistent with the study's purpose of assessing promotional readiness on the basis of employee performance and work-related measures instead of static profile characteristics. The SHAP-based interpretation also supports this conclusion by showing that Performance KPI had the highest contribution, while Profile KPI had the lowest contribution.

Thus, the KPI audit framework presented can be used to identify and evaluate the relationship between measurable employee performance and historical promotion decisions as a diagnostic tool. The framework does not replace human judgement. Instead, it helps identify potential inconsistencies between measurable employee performance indicators and historical promotion outcomes, thereby supporting more transparent, consistent, and evidence-based HR review processes.

5. Discussion and Conclusion

This section discusses the findings of the study in relation to fairness, transparency, and AI-supported decision-making in Human Resources. It also presents the limitations of the study, suggestions for future research, and the final conclusion.

5.1 Discussion of Findings

The results of the study indicate that an AI-powered KPI audit framework can provide valuable insight into how measurable employee performance indicators compare with historical promotion decisions. The Random Forest model showed stronger alignment when tested on KPI-based synthetic datasets, but this alignment decreased when the model was applied to the full real HR dataset.

The improved results from the synthetic datasets indicate that promotion readiness can be represented more consistently when making promotion decisions around both metrics and performance activity as part of the decision-making. Performance KPIs such as performance score, number of projects completed, number of training hours completed, attendance score, work hours worked, and employee satisfaction will provide information that can be used to assess the individual employee's contribution to the business; their work discipline; their effort towards development; and their engagement within the organization. In this way, an AI powered KPI audit framework will provide a basis for evaluating promotion readiness.

The fact that the alignment between historical and real-world HR decisions is less than expected in the historical data suggests a need for careful thought with respect to the implications for current practices. The lower than expected alignment does not imply that decisions made in the past were wrong or unfair, but it does suggest that actual HR decisions tend to be influenced by many factors that are outside of the stated criteria and outside of what is captured in the HR dataset. Promotion decisions are rarely based solely on numerical performance indicators; they can also be influenced by attributes such as leadership potential, communication and problem-solving ability, teamwork, manager's trust, contribution to strategic objectives, and perceived readiness for promotion.

In addition, organizational politics and informal decision-making processes may affect promotion results. For example, some employees may receive more support from their managers, have a greater presence in the organization, or have closer working relationships with decision-makers than others. Others may be affected by managerial preferences, department needs, succession planning, etc. Promotion decisions made based on the above attributes may create a gap between measurable performance and promotion results.

Mismatch cases identified by the audit framework are not necessarily evidence of discrimination but are more likely to signal that additional investigation is needed into the circumstances surrounding the promotion of employees classified as "hidden talent". Employees classified as hidden talent were considered ready for promotion according to the data from the original analysis, but have not been promoted in historical records.

There are multiple reasons why a hidden talent employee may not have been promoted, including having demonstrated high levels of measurable performance, lack of visibility, leadership recognition, communication abilities, or sufficient role availability at the time of the promotion decision. From an HR perspective, it is important to pay close attention to this group because high-performing employees may become demotivated by being overlooked for promotion, which will adversely affect their morale, erode organizational trust, and increase the risk of retention.

Cases of under- or low-merit promotion are also subject to careful interpretation. By definition, these are cases where an employee was promoted in fact, but was not identified as being ready for promotion based on their comparative performance level as reflected in the selected KPI indicators. While many under- or low-merit promotion cases include external factors (e.g., leadership potential), other cases could indicate inconsistency in the application of promotion criteria where an employee was promoted and the rationale for the promotion cannot be substantiated through both quantitative and qualitative measures.

The supplemental analysis of raw-feature importance, strategic KPI groups, and interpretation using SHAP is consistent with the goal of the proposed framework to promote fairness. The model relied predominantly on work-related and performance-based KPIs rather than static characteristics of the employee's profile. The two strongest contributors to promotion prediction were the Performance KPI and the Efficiency KPI with Profile KPI contributing the least. A fairness-based audit system should primarily focus on employee contribution and work-related KPIs rather than assessing the personal characteristics of the particular employee. However, even performance-based KPIs do have limitations in that they cannot measure qualitative attributes, such as a person's ability to lead, to demonstrate creativity, to manage a conflict, or to have an impact on other employees.

In summary, results indicate that AI can be leveraged as a diagnostic and audit tool in HR promotion processes. The model should not be considered an autonomous promotion decision-maker or as a definitive judgment on the merits of employees. Rather, it serves to substantiate an impartial basis for aiding HR professionals to identify situations that warrant further review. The importance of this human-in-the-loop approach is that it couples the consistency of data-driven analysis with the contextual awareness of human decision-makers. In this way, the proposed framework supports transparency, encourages evidence-based HR review, and helps organizations question whether their promotion decisions are aligned with measurable performance standards.

5.2 Limitations

Although this study generated valuable findings, several limitations should be considered when interpreting the results. First and foremost, the data for the study was drawn from a publicly available dataset consisting of employee performance and productivity data from Kaggle. Although the dataset is appropriate for developing and validating an AI-based audit framework, it does not capture the entire spectrum of complexities associated with the

promotion systems found in organizations across different industries. For instance, promotional decisions are influenced by various contextual factors such as specific factors relating to the industry of the company, the culture and style of the organization, the job role requirements, expectations of company leadership, team structures, internal HR policies, compensation structure and packages, succession planning, and strategic hiring. Thus, the promotion related information captured in the dataset reflects limited aspects of the context within which decisions about promotions are made in organizations. Hence, the results obtained in this study should not be extended to all organizations due to a lack of validation against actual corporate datasets.

Secondly, this dataset is primarily composed of measurable and quantifiable indicators of employee performance. The available variables include performance score, number of projects handled, training hours, work hours, overtime hours, attendance, employee satisfaction, years of employment, education level, and resignation status. That being said, the dataset does not include qualitative factors such as leadership potential, communication skills, creativity, teamwork, peer evaluation, managerial evaluation, conflict management skills, or organizational influence. Therefore, some of the mismatch cases identified by the model may have resulted from missing qualitative information rather than from unfair promotion decisions.

Thirdly, the Key Performance Indicator structures were created as part of this study. The KPIs identified (KPI Groups are: Performance KPI(s), Efficiency KPI(s), Commitment KPI(s), Profile KPI(s)) create an understandable framework for examining Promotion Readiness. However, different organizations may use different criteria to determine Promotion Readiness. A Sales Organization may have as their Promotion Criteria: Revenue Generation and Client Relationship Management; while a Technological Company may have as their Promotion Criteria: Technical Knowledge, Innovation, and Project Completion. Accordingly, it is necessary to be very careful when applying the grouping structure of these Key Performance Indicators to other types of organizations.

Fourth, a Random Forest classification model is employed in this study as the primary algorithm (predictor). While Random Forest is appropriate for analyzing tabular HR data and provides an output of feature importance, there exists a variety of other algorithms (i.e., Logistic Regression, Gradient Boosting, XGBoost, LightGBM, Support Vector Machines) which may yield different results from Random Forest for promotion audit analysis. As such, the current study's results should be perceived as the results of a single modelling method rather than generic conclusion of model results.

Fifth, the results of the study were supported with ROC-AUC analysis, 5-Fold Cross Validation, hyperparameter tuning via GridSearchCV, and explainability assessments via SHAP, however, there still exist certain limitations. The model was validated using the same dataset it was built from, and was never tested against other companies' datasets outside of the datasets the model was validated on. Thus, the results of this study should be taken as internal validation and not indicative of how the model will generalize across other companies. It

would be more rigorous for subsequent research to employ external validity, temporal validity and multiple-organizational datasets for repeat cross-validation.

Another limitation of the implementation of SHAP was that it was performed at the strategic KPI (key performance indicator) groups and only at that level of analysis. Thus, it was able to provide a global level of interpretability, but did not analyze how each of the original variables contributes to each of the individual employee's predictions of promotions through use of SHAP. For these reasons, the level of explainability of the model was much higher at the global and KPI levels than it was on an individual employee level.

Lastly, the intent of the designed framework is to serve as a retrospective audit mechanism in that it will conduct an analysis of the promotion outcomes that occurred after the decisions were made. Therefore, the framework was not capable of real-time decision support for promotions. To effectively function as a real-time decision-support system for promotional purposes, this framework would require integration with ethical AI governance, human review processes, data privacy measures, and standards of organizational accountability.

5.3 Future Research Suggestions

Further studies should examine the application of this framework to actual organization-level data in various industries with different types of organization structures. While this research was based on a publicly available employee-level performance dataset, validating these results against corporate-level data would enhance the external validity of the results. Using multi-company datasets may also allow for valuable insights about how promotion audit results based on key performance indicators differ by industry, company size, culture, and human resources management practices.

There are additional avenues for research, including the incorporation of some qualitative and behavioural variables. Examples of such qualitative variables include: Leadership evaluation, Peer feedback, Communication skills, Team contribution, Creativity, Conflict management, and Manager evaluation. Including these qualitative variables will provide a broader/complete picture of whether or not a person is ready for a promotion and/or whether or not a particular promotion decision is defensible in terms of qualitative factors not included in the present KPI structure.

Future research could include examinations of gender fairness and department-level fairness. This could provide a means for determining whether or not promotion mismatches are disproportionately skewed according to certain classifications such as gender, age, department, or job type. Research at a departmental level could be especially valuable as there can be varying criteria for promotions across different functional departments including sales, technology, finance, operations, and HR. Such subgroup analyses would help organizations understand whether fairness risks are general or concentrated in specific departments, job roles, or employee groups.

The model structure can be broadened by also comparing Random Forest to other Machine Learning methods. For example: Logistic Regression, Gradient Boosting, XGBoost, LightGBM, and Support Vector Machines. Additionally, future research can add more statistical methodology for validation through repeated cross-validation, time-based validations, threshold sensitivity analyses, and external validations on datasets acquired from other companies or organizations.

The framework can further deploy Explainable AI approaches in future iterations. The SHAP methodology was applied at the level of strategic KPI group, however, it would be beneficial for future research to test it at both raw feature and specific employee local explanation levels. As a result, HR professionals would have insight as to not only what KPI Groups are important in general, but also why an employee was determined to be promotion-ready/unready specifically. Additionally, integrating LIME and an interactive explanation panel would enhance transparency and build trust in AI based audit systems.

5.4 Conclusion

This study developed an AI-based KPI audit framework to assess the relationship between measurable employee performance indicators and historical promotion decisions. The purpose of developing the framework was not to create a means to automate promotion decisions but rather to develop a means to identify situations where measurable readiness indicators and actual organisational outcomes were not in alignment (mismatches).

The results showed that the Random Forest model achieved stronger alignment on KPI-based synthetic datasets, while this alignment decreased when the model was applied to the full real HR dataset. This suggests that historical promotion outcomes may not always be fully explained by measurable KPI-based performance criteria. This does not mean that there is proof of unfairness, but instead that there are audit signals indicating situations where a closer examination of HR practices is warranted.

Based on the audit framework developed in this study, three primary categories of audit results were established: Correctly Recognised, Hidden Talent, and Undeserved or Low Merit Promotions. These defined categories enhance the value of model outputs in helping HR understand how the model(s) can assist in identifying specific cases that require a more in-depth analysis for determining required action. In addition, HR may use the Hidden Talent category to help the organisation to identify employees who are, based on their measurable performance criteria, providing an indication of potential for advancement, although this indication was not clearly reflected in previous promotion decisions based on historical organisation outcomes.

The analysis of feature importance through additional raw features, key indicators, and SHAP methodology indicates that the model primarily focused on indicators related to performance and efficiency. This aligns with the study's overall goal of looking at fairness of readiness, since readiness was primarily assessed by using measurable job-related characteristics rather than fixed-profile characteristics. The study also acknowledges that

promotion decisions can be based on qualitative aspects that may not be adequately reflected in the dataset.

From a management perspective, the use of the proposed framework could allow for the HR review process to be more transparent and driven by evidence. Further, it may enable HR departments to examine whether promotion decisions are reflective of measurable performance criteria, identify employees who may have been passed over for promotions, and evaluate situations that may require stronger justification than is currently being provided. As a result, this framework can provide support for companies employing AI-based auditing systems rather than replacing management judgment.

This research shows the relevance of responsible AI governance within HR. The use of such AI tools must have defined lines of accountability along with human input, data protection rights, and proper interpretation of model outputs. A human being must support an algorithmic result rather than one replacing an organization's contextual judgement. With the responsible use of AI, HR processes can be more equitable through increased transparency, consistency, and reviewability in promotion processes.

In summary, the proposed AI audit framework demonstrates that machine learning can be used not only for prediction, but also for organizational fairness auditing. By comparing the measurable readiness indicators to historical outcomes, organizations can identify inconsistencies and thus implement more objective HR processes. Ultimately, the framework can be expanded into regular promotion audit processes, AI dashboards, and wider ethical HR governance practices within organizations.

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